2 Aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, are both used as reagents and catalysts.

(a) The melting point of Al_2O_3 is 2072 °C. The melting point of P_4O_{10} is 340 °C.

Explain the difference in the melting points of these two compounds.

.....

.....

.....

..... [3]

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH.

equation 1 $\quad CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3.

(i) Explain how the presence of a catalyst affects a chemical reaction.

.....

.....

..... [1]

3 **(a)** Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.

(i) State the compound that is heated with long-chain hydrocarbons to produce more useful smaller alkanes and alkenes.

..... [1]

(b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10}.

reaction 1 $C_4H_8 + H_2 \rightarrow C_4H_{10}$

(i) Name a catalyst for reaction 1.

..... [1]

(ii) Define activation energy, E_A.

.....

..... [1]

(iii) The Boltzmann distribution for the reaction mixture in reaction 1 is shown in Fig. 3.1.
Use the Boltzmann distribution to explain the effect of adding a catalyst on the rate of reaction.

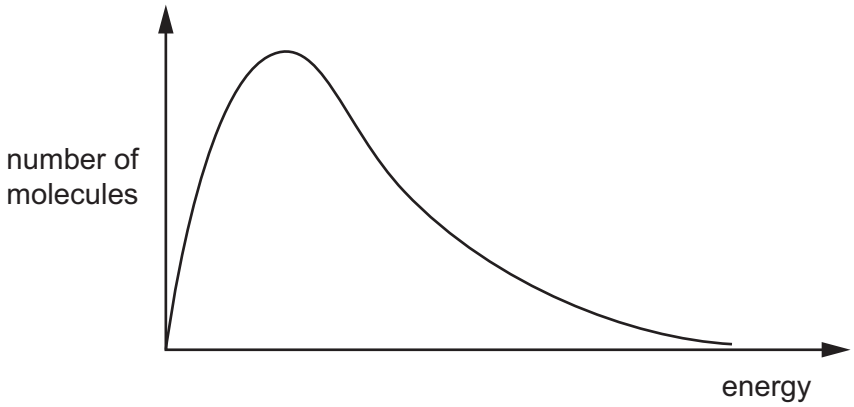


Fig. 3.1

.....

.....

.....

..... [2]

4 The reaction between $Na_2S_2O_3(aq)$ and $HCl(aq)$ is monitored at constant temperature.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$$

Fig. 4.1 shows how the concentration of $HCl(aq)$ varies with time.

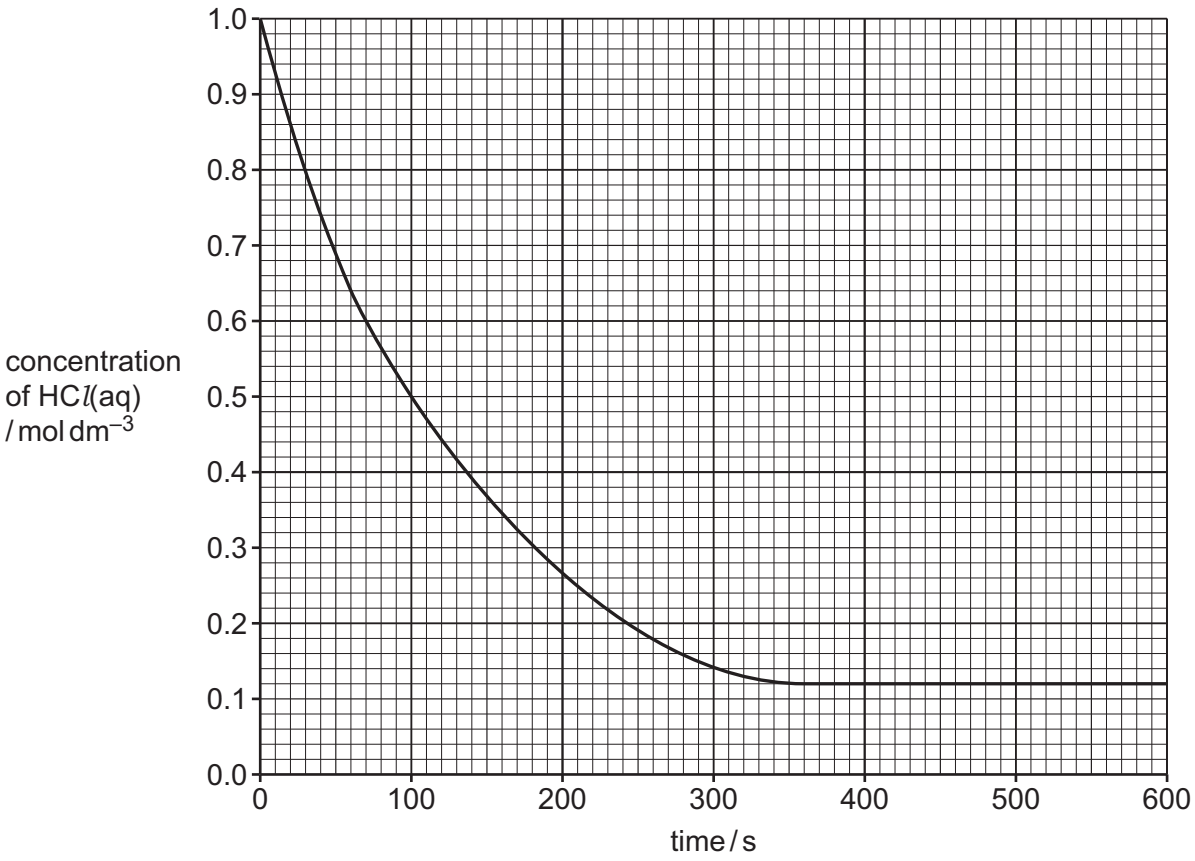


Fig. 4.1

(a) (i) Use Fig. 4.1 to find the average rate of change of concentration of $HCl(aq)$ in this reaction between 0–100 seconds and between 400–500 seconds. Include units in your answers.

0–100 seconds ..... units .....

400–500 seconds ..... units .....

[2]

(iii) Explain why the rate of reaction changes with time.

.....

.....

..... [1]

(v) Explain why the rate of reaction **cannot** be monitored accurately by measuring the volume of $SO_2(g)$ produced in this reaction.

.....

..... [1]

3 **(a)** C_3H_6 reacts with HBr(g) in an addition reaction.

(i) Define addition reaction.

.....

..... [1]

(b) H_2(g) and nickel are added to alkene **X**. Fig. 3.2 shows how the concentration of **X** changes with time.

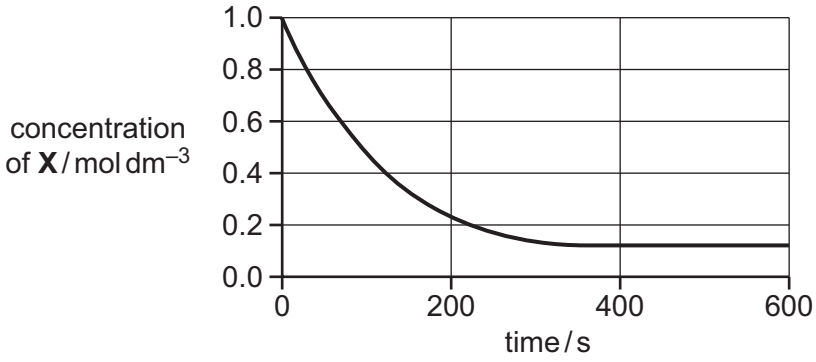


Fig. 3.2

(i) Use Fig. 3.2 to identify the limiting reagent in this reaction. Explain your answer.

.....

..... [1]

(ii) Use Fig. 3.2 to describe how the gradient changes as the reaction proceeds. State what this shows about the rate during this reaction.

.....

.....

..... [1]

(d) Hydrocarbon **Y** reacts with $H_2(g)$.

Fig. 3.3 shows the distribution of energies of $H_2(g)$ at temperature T.

Area **A** represents the number of molecules with energy greater than or equal to the activation energy, E_A, at temperature T.

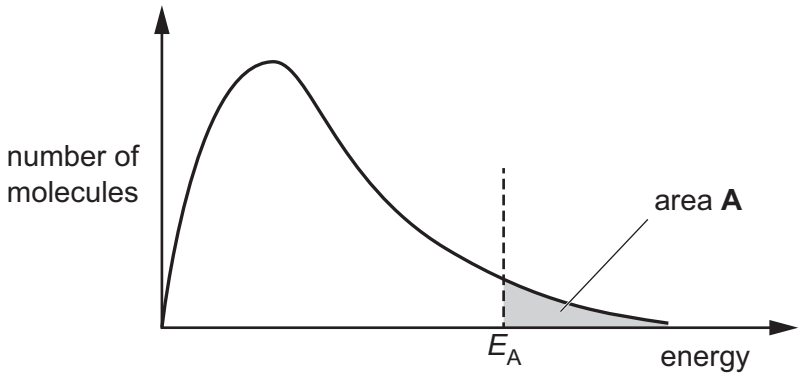


Fig. 3.3

(i) Annotate Fig. 3.3 to show the effect of adding nickel to **Y** and $H_2(g)$ at temperature T. [1]

(ii) Area **B** (**not** labelled on Fig. 3.3) represents the number of molecules with energy greater than or equal to the activation energy when nickel is added at temperature T.

State the difference, if any, between areas **A** and **B**. Explain the significance of your answer on the rate of hydrogenation of **Y**. Give your answer in terms of collisions.

.....

.....

..... [2]

3 **(a)** Hexene reacts with hydrogen gas to produce hexane.

reaction 1 $C_6H_{12}(l) + H_2(g) \rightarrow C_6H_{14}(l)$

Fig. 3.1 shows the distribution of energies of $H_2(g)$ molecules at temperature *T*.

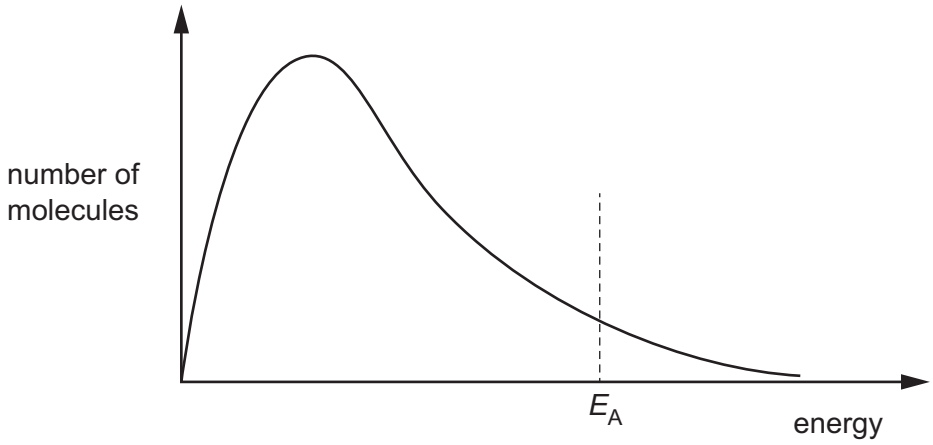


Fig. 3.1

(i) Sketch on Fig. 3.1 the shape of the curve for the same sample of $H_2(g)$ molecules when the temperature is increased. [2]

(ii) Explain why increasing the temperature increases the rate of reaction 1.

.....

.....

..... [2]

(iii) State the role of nickel when it is added to reaction 1.

..... [1]

(iv) Annotate Fig. 3.2 to show the effect of adding nickel to reaction 1 at temperature *T*.

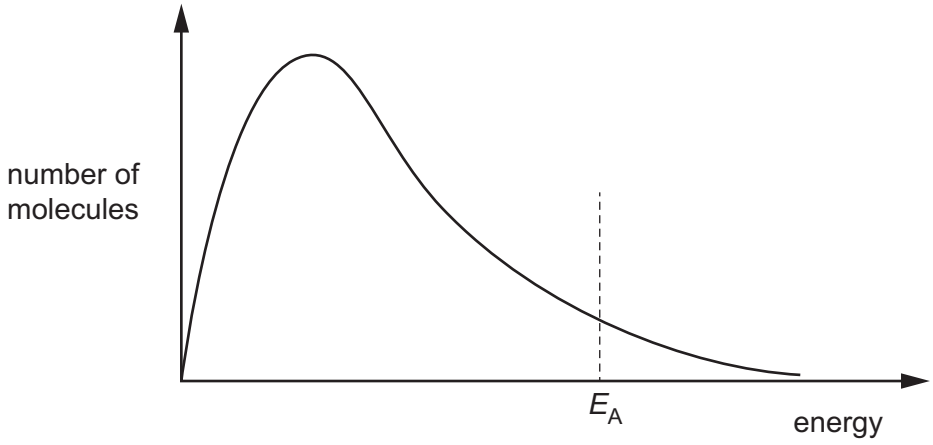


Fig. 3.2

[1]

2 $N_2(g)$ reacts with $H_2(g)$ in the Haber process, as shown in reaction 1.

reaction 1 $\quad N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -x\ kJ\,mol^{-1}$

Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	**B**	**C**
initial molar ratio of N_2 : H_2 added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $NH_3(g)$ at equilibrium	58	x	y

(a) Describe and explain the change, if any, to the percentage yield of $NH_3(g)$ produced in **B** compared to **A**.

.....

.....

..... [1]

(b) **(i)** Describe and explain the change, if any, to the percentage yield of $NH_3(g)$ produced in **C** compared to **A**.

.....

.....

..... [1]

(ii) Describe and explain the change to the rate of the forward reaction that occurs to establish the equilibrium in **C** compared to **A**.

You do **not** need to refer to the Boltzmann distribution in your answer.

.....

.....

.....

..... [2]

3 Nitrogen, N_2, is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1 .....

2 .....

[2]

(c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3.

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92\,kJ\,mol^{-1}$$

(i) Use the information in **(c)** to complete Table 3.1.

Table 3.1

compound	enthalpy change of formation, ΔH_f / kJ mol^{-1}
N_2	
H_2	
NH_3	

[2]

(ii) Explain how the presence of a catalyst affects the reaction.

.....

.....

..... [1]

(iii) State and explain the effect, if any, on the **rate** of the Haber process as the pressure is lowered.

.....

.....

.....

..... [2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(b) $KClO_3(s)$ decomposes when heated.

$MnO_2(s)$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $MnO_2(s)$ on the decomposition of $KClO_3(s)$.

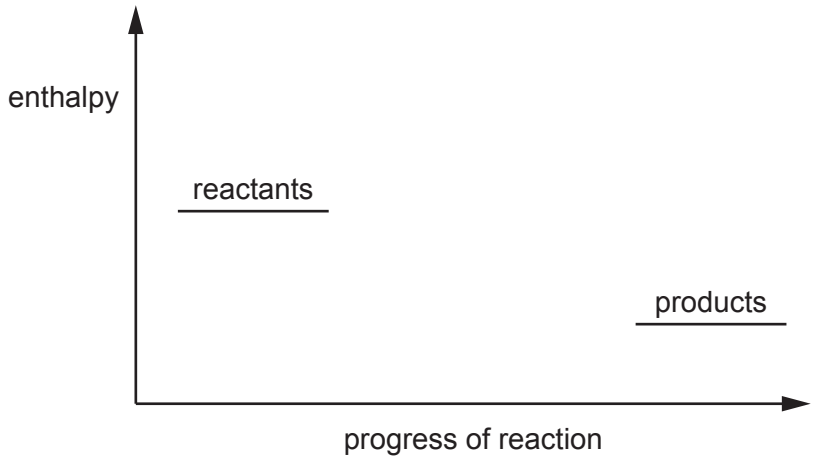


Fig. 3.1

[2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(b) $KClO_3(s)$ decomposes when heated.

$MnO_2(s)$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $MnO_2(s)$ on the decomposition of $KClO_3(s)$.

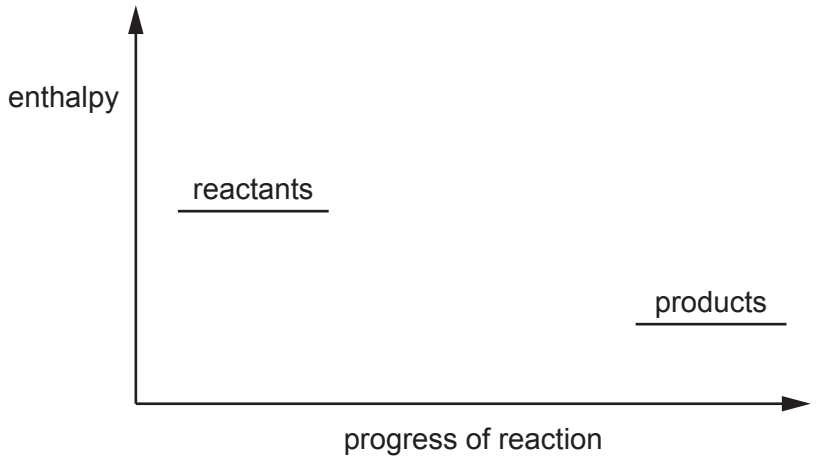


Fig. 3.1

[2]

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4.

reaction 1 $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r, for reaction 1.

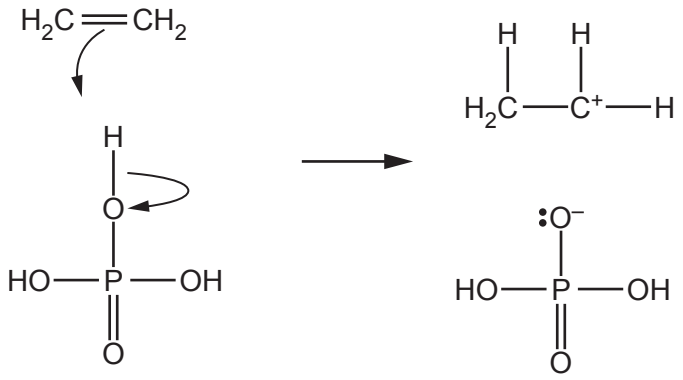
Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

ΔH_r = ... kJ mol^{-1} [2]

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

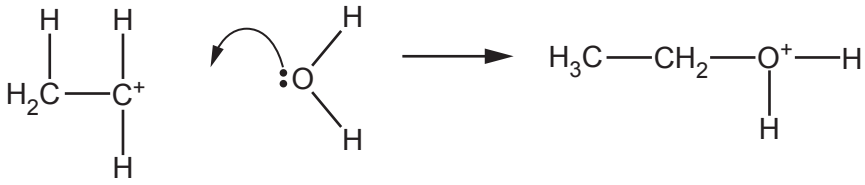


Fig. 4.1

(i) Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

.....

..... [1]

(iii) Complete Fig. 4.2 to show the mechanism for step 3 of reaction 1. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

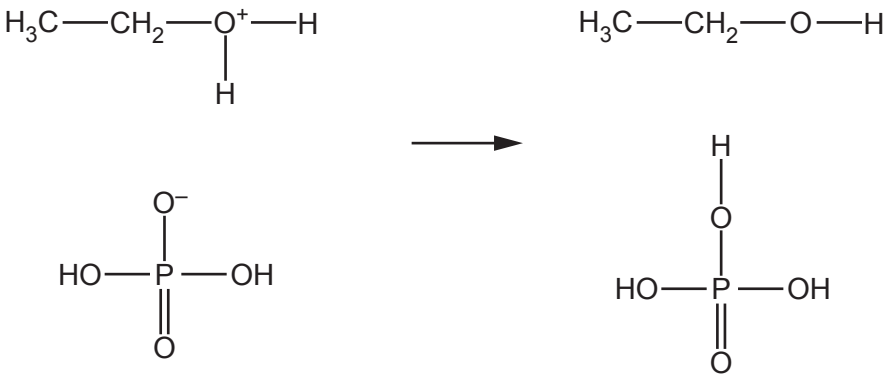


Fig. 4.2

[2]

(iv) Describe how a catalyst affects a reaction. Explain your answer.

.....

..... [2]

(v) Use Fig. 4.1 and Fig. 4.2 to justify why H_3PO_4 is described as a catalyst in reaction 1.

.....

..... [1]

5 **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

(i) Construct an equation for the complete combustion of **M**.

..... [2]

(b) When petrol is burned in an internal combustion engine, oxides of nitrogen are released into the atmosphere. Oxides of nitrogen are responsible for the formation of acid rain.

(i) Suggest the conditions required for the production of oxides of nitrogen during combustion of **M** in an internal combustion engine. Use an appropriate equation in your answer.

.....

.....

.....

..... [2]

(ii) Describe how acid rain is formed in the atmosphere in the presence of oxides of nitrogen and SO_2. Identify the role of the oxides of nitrogen in this process. Include **all** relevant equations.

.....

.....

.....

..... [3]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

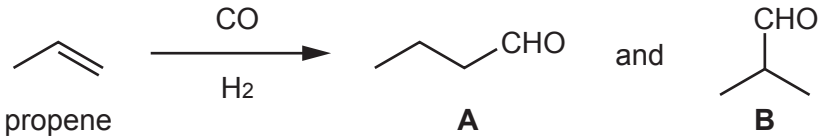


Fig. 3.1

(a) **(i)** Define addition reaction.

.....

..... [1]

(d) The reaction of ethene, C_2H_4, with a 1:1 mixture of CO and H_2 is shown in equation 1.

equation 1 $C_2H_4(g) + CO(g) + H_2(g) \rightleftharpoons CH_3CH_2CHO(g)$

At atmospheric pressure a cobalt-based catalyst is used in this reaction.

(i) State and explain the effect of using a catalyst on this reaction.

.....

.....

.....

..... [2]